# EXACT ACTION SEPARATION FOR A PRETZEL-TANGLE COMPOSITION IN THE PILLOWCASE

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Abstract. Let $P$ be the smooth stratum of the traceless SU(2) character variety of the four-punctured sphere. We study the immersed curve obtained by composing the rational-tangle arcs for $Q_{1/3}$ and $Q_{1/7}$ with the small-holonomy-perturbation correspondence associated to a three-strand tangle. After fixing an equivariant Liouville completion of $P$, we prove that the truncated correspondence is exact for every sufficiently small positive perturbation and that a half-period symmetry fixes the relative primitive on the two components of the output.

The singular composition consists of 21 separated graph sheets joined across six fold circles. Their twelve endpoints have nondegenerate hyperbolic smoothing germs. From the resulting exact path word we enumerate 58 seam-overlap candidates. Rational interval certificates for the normal-displacement equations retain exactly 50 transverse ordered self-intersection generators. An area-residue calculation evaluates every limiting action as a rational multiple of $\pi^2$: 42 are negative and 8 are positive, no action vanishes, and the smallest absolute limiting action is $\pi^2/42$. The census and all action signs persist for sufficiently small positive perturbation. The proof is accompanied by a pure-rational executable certificate. No holomorphic-quilt count, bounding cochain, or instanton-pillowcase comparison is asserted.

## 1. Introduction

The traceless SU(2) character variety of the four-punctured sphere is the pillowcase

$$P = (S^1 \times S^1)/(\gamma, \theta) \sim (-\gamma, -\theta).$$

Its smooth stratum $P^*$ is obtained by deleting the four orbifold points; we use the standard Liouville completion near those punctures [CHKK22, Appendix A.1.1]. Rational tangles give proper Lagrangian arcs $A_r \subset P^*$, while a holonomy perturbation of the three-strand tangle gives an immersed Lagrangian correspondence

$$C_t = C^{\mathrm{reg}}_{3,t} \looparrowright (P_1^*)^- \times (P_2^*)^- \times P_3^*$$

for small $t > 0$ [HK18, Smi24]. We consider the direct fiber product and its output immersion

$$\widetilde{L}_t = (A_{1/3} \times A_{1/7}) \times_{P_1 \times P_2} C_t, \qquad L_t = \pi_3(\widetilde{L}_t) \looparrowright P_3^*. \tag{1}$$

The pair of slopes $1/3$ and $1/7$ is the character-variety composition associated to the pretzel-tangle presentation of $P(-2, 3, 7)$ used in the companion calculation [Wue26].

The purpose of this paper is finite and geometric. It determines the exact immersed-curve action data of (1); it does not construct any moduli space of figure-eight bubbles. This distinction matters because strip shrinking can produce Morsified trees of several holomorphic components [BW18]. A list of candidate output generators is not a count of those trees.

We fix the following terminology. If a transverse self-intersection of $L_t$ has two preimages $z_-$ and $z_+$, the orientation convention for immersed Floer theory selects one ordered pair $x = (z_-, z_+)$

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in degree one. We call this ordered generator a *root*. If $f_t$ is the primitive inherited from the exact fiber product, its action is

$$\mathcal{A}_t(x) = f_t(z_-) - f_t(z_+). \tag{2}$$

The word "root" in this paper therefore means an ordered branch jump, not a zero of an unspecified function. The *short deck class* of a root is the reduced torus-deck displacement of the ordinary closing loop used in the area-residue calculation; for an odd seam separation, the construction first doubles the path and then removes that artificial factor of two.

**Theorem 1.1** (Exact action census)**.** *There is an equivariant Liouville completion of the pillowcase and a number $t_0 > 0$ with the following properties for every $0 < t < t_0$.*

1. *The compactly truncated correspondence $C_t$ is exact. The direct output $L_t$ inherits a primitive whose relative normalization across its two open parameter arcs is fixed by a half-period symmetry.*
2. *The curve $L_t$ has exactly 50 transverse degree-one roots arising from the singular composition: 28 diagonal-circle roots and 22 circle-circle roots. They occur in 25 half-period pairs.*
3. *The limiting actions as $t \to 0^+$ are nonzero rational multiples of $\pi^2$. Exactly 42 roots have negative limiting action and 8 have positive limiting action. The minimum absolute limiting action is $\pi^2/42$, and all 50 action signs persist throughout $0 < t < t_0$.*

*The four positive half-period representatives have short deck classes and limiting actions*

| *seam pair* | *short deck* | $\lim_{t\to 0^+} \mathcal{A}_t/\pi^2$ |
|---|---|---|
| $(1,7)$ | $(3,1)$ | $3/7$ |
| $(2,7)$ | $(8,4)$ | $5/14$ |
| $(2,17)$ | $(3,1)$ | $3/14$ |
| $(3,4)$ | $(10,5)$ | $5/42.$ |

Three ingredients make the statement exact. First, the Liouville primitive is normalized equivariantly and its periods are checked on the full singular fiber, including seam-born and corner-link cycles. Hamiltonian-composition flux then transports this vanishing to $t > 0$. Second, the singular output is recorded as an exact piecewise-linear word. Its only failures of the graph description occur at six explicitly determined stabilizer circles, whose endpoints carry hyperbolic smoothing germs. Third, all circle-circle intersections reduce to quadratic equations over $\mathbb{Q}(2\cos(\pi/7))$. Rational enclosures certify the physical roots and their derivative signs. The action of each surviving root is then a signed area minus a lattice-residue term.

Section 2 fixes the primitive and proves exactness. Section 3 describes the six folds, the hyperbolic smoothing, and the 21 separated graph sheets. Section 4 proves the exact root census and action table. Section 5 records the precise energy consequence and the boundary of the result.

## 2. Exact primitives and half-period normalization

Write $(\gamma, \theta)$ for the torus-cover coordinates on the pillowcase. On the central cylinder

$$U = (0, \pi) \times S^1 \subset P^*,$$

the Goldman form is $d\gamma \wedge d\theta$. The choice of a global primitive on the completed four-punctured sphere includes residue data at its four ends.

**Lemma 2.1** (Normalized pillowcase completion)**.** *Let*

$$\tau[\gamma, \theta] = [\gamma, \theta + \pi], \qquad \sigma[\gamma, \theta] = [\pi - \gamma, -\theta].$$

*There is a $\langle\tau,\sigma\rangle$-invariant corner truncation $P^\delta \subset P^*$ and a $\langle\tau,\sigma\rangle$-invariant Liouville completion $(\widehat{P}^\delta, \lambda)$ such that*

$$d\lambda = d\gamma \wedge d\theta \quad \text{on } P^\delta, \qquad \lambda = (\gamma - \pi/2)\, d\theta$$

*near the circle $\{\gamma = \pi/2\}$. The four oriented end residues of $\lambda$ are equal.*

*Proof.* Choose the four deleted disks equivariantly. Since the compact surface with boundary $P^\delta$ has zero second cohomology, the restricted Goldman form has a primitive $\lambda'$. On an annular neighborhood of $\{\gamma = \pi/2\}$, add a closed one-form to kill the period of $\lambda' - (\gamma - \pi/2)d\theta$, and then subtract the differential of a cutoff function to make the difference vanish there. Average the construction over $\langle\tau,\sigma\rangle$. The group acts transitively on the corners, so the four oriented boundary periods agree.

Adjust the primitive in disjoint boundary collars so that its Liouville vector field is outward transverse, and attach the positive symplectizations of the boundary circles. These adjustments can be made equivariantly and do not change the primitive on the central annulus. □

Put

$$\Lambda_P = -\lambda_1 - \lambda_2 + \lambda_3.$$

On a smooth torus-cover chart of the unperturbed correspondence, Smith's boundary coordinates are

$$\gamma_1 = \gamma_2 = \gamma_3 = \gamma, \qquad \theta_1 = \beta \sin\alpha, \qquad \theta_2 = \theta - \beta\sin\alpha, \qquad \theta_3 = \theta \tag{3}$$

[Smi24, Eqs. (3.1.4)–(3.1.5)]. At $\gamma = \pi/2$, the normalized local primitive therefore satisfies $\Lambda_P|_{C_0} = 0$.

**Proposition 2.2** (Exactness of the small perturbation)**.** *For a generic symmetric truncation $P^\delta$ and every sufficiently small $t > 0$,*

$$[\ell_t^* \Lambda_P] = 0 \quad \text{in } H^1(C_t^\delta; \mathbb{R}),$$

*where $\ell_t : C_t^\delta \looparrowright (P^\delta)^- \times (P^\delta)^- \times P^\delta$ is the restriction immersion. The assertion includes the cycles created when Smith's singular seam is smoothed and all corner-link cycles.*

*Proof.* We first calculate on the full singular parameter fiber. Smith's regular parameter space is a two-fold cover of $C_0$ and decomposes as

$$\mathcal{V}_0^* \cong \left[S^2 \times (S^1 \setminus S^0)\right] \sqcup \left[T^2 \times (S^1 \setminus S^0)\right]$$

[Smi24, Def. 3.6]. The first summand has no first real cohomology. On the second, represent the two torus generators at $\gamma = \pi/2$. Equation (3) gives angular increments $(2\pi, -2\pi, 0)$ for the $\beta$-circle and $(0, 2\pi, 2\pi)$ for the $\theta$-circle. The normalized primitive vanishes on both generators. Thus the period class vanishes on the regular part.

It remains to include cycles through the singular seam. The full zero set is

$$\mathcal{V}_0 = A \cup H_0 \cup H_\pi, \qquad A = \{\alpha = \pi/2\} \cong T^3, \qquad H_0 \sqcup H_\pi = \{\gamma = 0, \pi\} \cong S^0 \times S^1 \times S^2.$$

Cut $A$ along the four subtori $\beta = \epsilon\pi$ and $\theta - \beta = \epsilon\pi$, $\epsilon \in \{0, 1\}$. Mayer-Vietoris gives a spanning set for first homology consisting of the angular directions on these subtori, four $\gamma$-circles, and the two directions on every corner link. The angular periods vanish by the preceding paragraph.

Orient the four pillowcase meridians positively and use $(m_{0\pi}, m_{\pi 0}, m_{\pi\pi})$ as a basis. If $\mathcal{A}_\delta = \int_{P^\delta} d\lambda$, direct substitution in the boundary formulas gives

$$\begin{array}{c|c|c} c & [-p_1(c) - p_2(c) + p_3(c)] & \displaystyle\int_c r_0^* \Lambda_P^{(00)} \\ \hline A_0^\gamma & (0,0,0) & 0 \\ A_\pi^\gamma & (2,0,2) & \mathcal{A}_\delta \\ B_0^\gamma & (0,1,-1) & 0 \\ B_\pi^\gamma & (-2,-1,-1) & -\mathcal{A}_\delta. \end{array} \tag{4}$$

Here $\lambda^{(00)}$ is a primitive with meridian periods $(-\mathcal{A}_\delta, 0, 0, 0)$. The symmetric primitive has period $-\mathcal{A}_\delta/4$ on each meridian. Pairing the difference with the middle column changes the final column by $0, -\mathcal{A}_\delta, 0, \mathcal{A}_\delta$, respectively, so all four $\gamma$-periods vanish.

At a parameter corner $(g, h, k) \in \{0,1\}^3$, an oriented basis $(u, v)$ of its torus link has meridian winding triples

$$\begin{array}{c|c|c} (g,h) & (w_1,w_2,w_3)(u) & (w_1,w_2,w_3)(v) \\ \hline (0,0) & (-1,1,0) & (1,1,2) \\ (0,1) & (-1,-1,-2) & (-1,1,0) \\ (1,0) & (1,1,2) & (-1,1,0) \\ (1,1) & (1,-1,0) & (1,1,2). \end{array} \tag{5}$$

Every vector satisfies $w_3 = w_1 + w_2$. Since the four end residues agree, the signed period of a link circle is a common multiple of $-w_1 - w_2 + w_3$ and hence vanishes. We have therefore proved that $r_0^* \Lambda_P$ has zero period on every class in $H_1(\mathcal{V}_0^\delta; \mathbb{R})$.

For $t > 0$, cut out the perturbation solid torus. The perturbed correspondence is a geometric composition of a fixed restriction immersion with the solid-torus arc

$$[e^{is}] \longmapsto [e^{is}, e^{if_t(s)}],$$

which is a Hamiltonian deformation [CHK22, Eq. (6-4) and Sec. 6B]. If $v_t$ is the velocity of the composed family, the Lagrangian condition gives

$$\ell_t^* \iota_{v_t}(-\omega_1 - \omega_2 + \omega_3) = -d(H_t \circ \mathrm{pr}), \tag{6}$$

with the convention $\iota_{X_{H_t}}\omega = -dH_t$. Hence $\ell^*\Lambda_P + G\,dt$ is closed on the smooth trace of the family.

Let $q_t : \mathcal{V}_t^\delta \to C_t^\delta$ be Smith's two-fold parameter cover. The closure of its trace over $0 \le t \le \epsilon$ is a compact subanalytic pair with the zero fiber, so it admits a compatible triangulation. After decreasing $\epsilon$, a regular neighborhood of the zero fiber contains the trace and retracts onto that fiber. Thus every one-cycle in a positive fiber is joined by a singular two-chain to a cycle in the full zero set; vanishing cycles are included. The Hamiltonian graph and all boundary maps extend analytically to $t = 0$, so the trace form extends continuously and is smooth on each stratum of the triangulation. Stokes' theorem applied stratum by stratum gives

$$\int_{\widetilde{a}_t} q_t^* \ell_t^* \Lambda_P = \int_{a_0} r_0^* \Lambda_P = 0.$$

Pullback on first real cohomology under a finite cover is injective. This proves the proposition. $\square$

The rational arcs are contractible and can be normalized equivariantly, so $A_{1/3} \times A_{1/7}$ is exact. Primitives add under transverse geometric composition. Proposition 2.2 therefore gives a primitive on every component of the direct fiber product (1). The output, however, passes through two deleted corners in each lifted period. Its two open arcs initially have independent additive constants; the next proposition fixes their relative normalization.

**Proposition 2.3** (Half-period normalization)**.** *On the parameter cover of the direct composition, all-corner points on the main lift are*

$$z_j = (\gamma, \theta, \alpha, \beta) = (21j\pi, 10j\pi, \pi/2, 7j\pi), \qquad j \in \mathbb{Z}.$$

*Removing two consecutive corner points cuts a lifted period into two open arcs. Define*

$$T(\gamma, \theta, \alpha, \beta) = (\gamma + 21\pi, \theta + 10\pi, \alpha, \beta + 7\pi).$$

*Modulo the parameter period lattice, $T$ is an involution exchanging the two arcs. With $\rho = \tau\sigma$, the boundary maps satisfy*

$$r_1 T = \rho r_1, \qquad r_2 T = \rho r_2, \qquad r_3 T = \sigma r_3. \tag{7}$$

*The inherited primitive $F$ may be normalized so that $F \circ T = F$.*

*Let $I_0$ be the open arc $0 < \gamma < 21\pi$, let $I_1 = T(I_0)$, and define*

$$\operatorname{can}(z) = z \quad (z \in I_0), \qquad \operatorname{can}(z) = T^{-1}z \quad (z \in I_1).$$

*For every root $x = (z_-, z_+)$, including a root whose preimages lie on different open arcs,*

$$\mathcal{A}_t(x) = \int_{\operatorname{can}(z_+)}^{\operatorname{can}(z_-)} r_3^*\lambda, \tag{8}$$

*where the integral follows the oriented subarc of $I_0$.*

*Proof.* At a parameter corner $(g, h, k)$, the three boundary characters are $(g, k)$, $(g, k+h)$, and $(g, h)$. For odd $q$, a corner $(g, r)$ lies on $A_{1/q}$ precisely when $r = g$. The two input equations therefore give $k = g$ and $h = 0$, which yields exactly the displayed corner lifts. Their consecutive spacing shows that each full period contains two corner cuts.

The square of $T$ is the parameter period $(42\pi, 20\pi, 0, 14\pi)$. Direct substitution in the defining traceless equation shows that $T$ preserves the parameter space, and substitution in the boundary words gives (7). The map $\rho$ preserves both odd-slope arcs. Average the primitive on each rational arc under $\rho$ and the primitive on $C_t$ under $T$. The normalized Liouville form is invariant under $\tau$ and $\sigma$, so the averages have the same exterior derivatives and the composition primitive satisfies $F \circ T = F$.

On either open component, $dF = r_3^*\lambda$. Replace both endpoints by their representatives in $I_0$ and integrate from the positive endpoint to the negative endpoint. The interior of $I_0$ contains no all-corner point, so no residue detour is involved and (8) follows. □

## 3. The six folds and the separated graph sheets

The partial composition

$$F_t = A_{1/3} \times_{P_1} C_t \looparrowright P_2^- \times P_3$$

is used here only to describe the finite geometry of the direct output. No associativity or curved-composition theorem is invoked.

**Proposition 3.1** (Six-circle and fold table)**.** *The singular partial composition has six stabilizer circles. Their $P_2$ seam values, the corresponding $A_{1/3}$ angle $\theta_1$, and the image interval $[\ell_i, h_i]$ in the third pillowcase are*

| $i$ | $(\gamma_2, \theta_2)$ | $\theta_1$ | $[\ell_i, h_i]$ |
|---|---|---|---|
| 1 | $(0, 2\pi/7)$ | $2\pi/3$ | $[8\pi/21, 20\pi/21]$ |
| 2 | $(0, 4\pi/7)$ | $2\pi/3$ | $[2\pi/21, 16\pi/21]$ |
| 3 | $(0, 6\pi/7)$ | $2\pi/3$ | $[4\pi/21, 10\pi/21]$ |
| 4 | $(\pi, \pi/7)$ | $\pi/3$ | $[4\pi/21, 10\pi/21]$ |
| 5 | $(\pi, 3\pi/7)$ | $\pi/3$ | $[2\pi/21, 16\pi/21]$ |
| 6 | $(\pi, 5\pi/7)$ | $\pi/3$ | $[8\pi/21, 20\pi/21]$. |

*If $\psi$ parametrizes a stabilizer circle, then*

$$\cos\theta_3(\psi) = \cos\theta_1\cos\theta_2 + \sin\theta_1\sin\theta_2\cos\psi. \tag{9}$$

*Both endpoints of every interval are nondegenerate folds. The pillowcase symmetry pairs rows* $(1,6)$, $(2,5)$, *and* $(3,4)$, *including their fold coefficients.*

*Proof.* A lift of $A_{1/q}$ satisfies $q\theta - \gamma \in 2\pi\mathbb{Z}$. Its noncorner seam values are

$$A_{1/3}: \ (0, 2\pi/3), \ (\pi, \pi/3)$$

and

$$A_{1/7}: \ (0, 2\pi k/7), \ (\pi, (2k-1)\pi/7), \qquad k = 1,2,3.$$

Pairing the unique $A_{1/3}$ value on each seam with the three $A_{1/7}$ values gives the six circles. Choose the common seam axis as polar axis. The Euclidean dot product of the two remaining axes gives (9), and hence

$$[\ell_i, h_i] = \big[|\theta_1 - \theta_2|, \pi - |\pi - (\theta_1+\theta_2)|\big].$$

At the two endpoints,

$$\theta_3(\psi) = \ell_i + \frac{\sin\theta_1\sin\theta_2}{2\sin\ell_i}\psi^2 + O(\psi^4),$$
$$\theta_3(\pi+u) = h_i - \frac{\sin\theta_1\sin\theta_2}{2\sin h_i}u^2 + O(u^4).$$

All coefficients are positive and finite. The symmetry assertion follows by replacing the two input angles by their complements. $\square$

**Proposition 3.2** (Hyperbolic smoothing at the twelve endpoints)**.** *Use the group-presentation convention*

$$p = \exp\big(t\ \mathrm{Im}(ac^{-1}x)\big), \qquad \Phi_t = \mathrm{Re}(xp^{-1}a^{-1}pb),$$

*with*

$$a = \mathbf{i}, \quad b = e^{\gamma\mathbf{k}}\mathbf{i}, \quad c = e^{\theta\mathbf{k}}\mathbf{i}, \quad x = \sin\alpha\cos\beta\,\mathbf{i} + \sin\alpha\sin\beta\,\mathbf{j} + \cos\alpha\,\mathbf{k}.$$

*Near the lower and upper endpoints of the ith circle there are smooth coordinates* $(u,v)$ *in which*

$$\{\Phi_t = 0\} = \{uv = \tau_i^{\pm}(t)\}, \qquad \tau_i^{\pm}(0) = 0, \qquad (\tau_i^{\pm})'(0) = \mp 2\sin\theta_1\sin\theta_2.$$

*Thus all twelve endpoints have nondegenerate hyperbolic smoothing germs, with opposite sectors at the two ends of each circle and the same three symmetry pairs as above.*

*Proof.* At $t = 0$ the defining equation is

$$\Phi_0(\gamma,\theta,\alpha,\beta) = \sin\gamma\cos\alpha.$$

Write $X = \gamma - \gamma_i$ and $Y = \alpha - \pi/2$, where $\gamma_i \in \{0,\pi\}$ and $\epsilon_i = \cos\gamma_i$. Quaternion multiplication gives

$$\Phi_t = \epsilon_i\big(-XY - 2t\sin\beta\sin(\theta-\beta)\big) + O\big(t(|X|+|Y|) + t^2 + (|X|+|Y|)^3\big)\,.$$

The Hessian in $(X,Y)$ is nonsingular with index one. The parametric Morse lemma therefore gives $uv = \tau(t)$ and

$$\tau'(0) = -2\sin\beta\sin(\theta-\beta).$$

At the lower and upper lifts the products are respectively $-\sin\theta_1\sin\theta_2$ and $\sin\theta_1\sin\theta_2$, proving the formulas. $\square$

**Proposition 3.3** (Off-seam graph sheets)**.** *Away from neighborhoods of the six seam values, every branch of $F_0$ is locally the graph of the symplectic shear*

$$\phi_c(\gamma, \eta) = (\gamma, \eta + \gamma/3 + c).$$

*On a torus-cover chart, the 21 local lifts of the direct composition are parallel and pairwise disjoint. Both statements persist on compact off-seam subsets for all sufficiently small $t > 0$.*

*Proof.* On the unperturbed main stratum, the three boundary coordinates are

$$p_1 = (\gamma, \beta), \qquad p_2 = (\gamma, \theta - \beta), \qquad p_3 = (\gamma, \theta).$$

The first rational arc gives $\beta = \gamma/3 + c$. With $\eta = \theta - \beta$, this is the displayed shear, and $d\gamma \wedge d(\eta + \gamma/3 + c) = d\gamma \wedge d\eta$. Smith's cross-resolution theorem confines every failure of this graph description to the six seam neighborhoods [Smi24, Thm. 4.22].

Choose lifts

$$\beta = \frac{\gamma + 2\pi k}{3}, \qquad \eta = \frac{\gamma + 2\pi \ell}{7}, \qquad (k, \ell) \in \mathbb{Z}/3 \times \mathbb{Z}/7.$$

Then

$$\theta = \frac{10}{21}\gamma + \frac{2\pi(7k + 3\ell)}{21}.$$

The homomorphism $(k, \ell) \mapsto 7k + 3\ell$ from $\mathbb{Z}/3 \times \mathbb{Z}/7$ to $\mathbb{Z}/21$ is bijective, so the 21 lifts have distinct intercepts. A compact off-seam set has positive minimum separation, which persists under the $C^1$ small-$t$ deformation. $\square$

## 4. The exact root-action table

Use scaled coordinates on the canonical arc $I_0$:

$$s = \frac{\gamma}{\pi}, \qquad n = \frac{21\theta}{\pi}.$$

A vertical torus period changes $n$ by 42, and the arc runs from $(0, 0)$ to $(21, 210)$.

**Lemma 4.1** (Singular first-arc word)**.** *Between consecutive integral values of $s$, the singular first arc has slope $d\theta/d\gamma = 10/21$. Its twelve signed vertical jumps are*

| $s$ | $i$ | $\Delta n$ | *$n$-interval* | $s$ | $i$ | $\Delta n$ | *$n$-interval* |
|---|---|---|---|---|---|---|---|
| 1 | 4 | $-6$ | $[4, 10]$ | 11 | 5 | 14 | $[110, 124]$ |
| 4 | 1 | $-12$ | $[22, 34]$ | 12 | 1 | 12 | $[134, 146]$ |
| 5 | 4 | 6 | $[32, 38]$ | 14 | 2 | $-14$ | $[152, 166]$ |
| 7 | 5 | $-14$ | $[44, 58]$ | 16 | 3 | 6 | $[172, 178]$ |
| 9 | 6 | 12 | $[64, 76]$ | 17 | 6 | $-12$ | $[176, 188]$ |
| 10 | 2 | 14 | $[86, 100]$ | 20 | 3 | $-6$ | $[200, 206]$. |

*At the other interior integral seams, the diagonal sheet has the eight points*

$$(2, 14), (3, 24), (6, 48), (8, 54), (13, 156), (15, 162), (18, 186), (19, 196).$$

*Proof.* Proposition 3.3 gives $\theta = \gamma/3 + \gamma/7 + c$, so increasing $s$ by one increases $n$ by ten. At an integral seam, substitute the six intervals of Proposition 3.1 and choose the lift whose incoming endpoint agrees with the preceding diagonal segment. Continuing through the outgoing endpoint gives the twelve jumps. The recurrence

$$n_j^- = n_{j-1}^+ + 10, \qquad n_j^+ = n_j^- + \Delta n_j$$

gives all displayed intervals and point values. The jumps sum to zero, so the endpoint is $(21, 210)$. $\square$

**Lemma 4.2** (Algebraic normal roots)**.** *Put $x=\cos\theta$ and $u=2\cos(\pi/7)$, so*

$$u^3-u^2-2u+1=0.$$

*On row $i$ of Proposition 3.1, the small-$t$ normal displacement is*

$$\gamma=j\pi+tG_i(\theta)+O(t^2),\qquad G_i(\theta)=\frac{2(a_i-\cos\theta)}{\cos\alpha_i(\theta)},\tag{10}$$

*where*

$$\cos^2\alpha_i(\theta)=\frac{b_i^2-(\cos\theta-a_i)^2}{\sin^2\theta},\qquad \operatorname{sgn}(\cos\alpha_i)=\begin{cases}+1, & i=1,2,3,\\ -1, & i=4,5,6,\end{cases}\tag{11}$$

*and*

$$a_1=a_6=\frac{2-u^2}{4},\qquad a_2=a_5=\frac{u^2-u-1}{4},\qquad a_3=a_4=\frac{u}{4},$$
$$b_i^2=\frac{3}{4}(1-4a_i^2).$$

*The physical simple solutions of $G_i=G_j$ on $0<\theta<\pi$ are exactly*

| *root* | *row pairs* $\{i,j\}$ | $x$*-box* | $21\theta/\pi$*-box* |
|---|---|---|---|
| A | $\{1,4\},\{3,6\}$ | $[0.178447, 0.178449]$ | $[9.300, 9.301]$ |
| B | $\{1,5\},\{2,6\}$ | $[-0.123491, -0.123489]$ | $[11.327, 11.328]$ |
| C | $\{1,6\}$ | $[-0.311746, -0.311743]$ | $[12.619, 12.620]$ |
| D | $\{2,3\},\{4,5\}$ | $[0.722519, 0.722522]$ | $[5.102, 5.103]$ |
| E | $\{2,4\},\{3,5\}$ | $[0.346009, 0.346012]$ | $[8.138, 8.139]$ |
| F | $\{2,5\}$ | $[0.111259, 0.111262]$ | $[9.754, 9.755]$ |
| G | $\{3,4\}$ | $[0.450483, 0.450486]$ | $[7.376, 7.377]$. |

*There is no physical root for the row pairs $\{1,2\},\{1,3\},\{4,6\},\{5,6\}$.*

*Proof.* At $t=0$, substitution of the seam rational-arc equations into the quaternionic boundary words gives (11). Differentiating the defining equation in the normal $\gamma$ direction and in $t$ gives

$$\partial_\gamma\Phi_0=(-1)^j\cos\alpha_i,\qquad \partial_t\Phi_0=-2(-1)^j(a_i-\cos\theta),$$

which proves (10). The formulas for $a_i$ and $b_i$ follow from the minimal polynomial of $u$.

For pairs of distinct symmetry types, squaring $G_i=G_j$ and clearing the positive fold denominators gives the three quadratics $P_{rs}(x)=A_{rs}x^2+B_{rs}x+C_{rs}$ with

$$(A_{12},B_{12},C_{12})=\left(u^2-2u-1,-\frac{5}{2}u^2+\frac{3}{2}u+\frac{5}{2},\frac{1}{4}u^2-\frac{1}{2}u-\frac{1}{4}\right),$$
$$(A_{13},B_{13},C_{13})=\left(u^2-\frac{1}{2}u-\frac{3}{2},-\frac{3}{4}u^2-\frac{1}{2}u+2,\frac{1}{4}u^2-\frac{1}{8}u-\frac{3}{8}\right),$$
$$(A_{23},B_{23},C_{23})=\left(u^2+u-2,u^2-\frac{5}{2}u-2,\frac{1}{4}u^2+\frac{1}{4}u-\frac{1}{2}\right).$$

The rational interval

$$1.801937735<u<1.801937737$$

contains $2\cos(\pi/7)$ and one simple zero of $u^3-u^2-2u+1$. Rational interval evaluation gives one physical root of $P_{12}$, one of $P_{13}$, and both roots of $P_{23}$ in the displayed boxes. Each polynomial has opposite strict signs at the endpoints of its box and a derivative of one strict sign there. Vieta's formula places the unused root of $P_{12}$ below $-1$ and that of $P_{13}$ above 1. The signs in (11) remove the extraneous squared solutions. For equal symmetry types the two functions agree only at $x=a_i$, producing C, F, G, and their derivatives are nonzero.

The angle boxes are also rationally certified. Machin's identity

$$\frac{\pi}{4} = 4\arctan\frac{1}{5} - \arctan\frac{1}{239}$$

and alternating sums give a rational $\pi$-interval of width less than $1.84 \times 10^{-35}$. Rational Taylor bounds for cosine then place every angle strictly inside its displayed box and certify the derivative signs. □

**Proposition 4.3** (Complete limiting action table)**.** *The path of Lemma 4.1 has* 58 *seam-overlap candidates. Exactly* 50 *persist as transverse degree-one roots for all sufficiently small $t > 0$:* 28 *are diagonal-circle roots and* 22 *are the simple circle-circle roots of Lemma 4.2. The eight excluded seam pairs are*

$$\{1,17\}, \{4,14\}, \{4,20\}, \{5,9\}, \{7,17\}, \{9,11\}, \{10,12\}, \{12,16\}.$$

*The following table gives one representative of each half-period pair. Here $d \in 42\mathbb{Z}$ is the offset in $n$, $r$ is a normal root or* pt *for a diagonal-circle root, $D$ is the short deck class defined above, and $o$ is the degree-one ordering. The columns $a, w, \kappa$ satisfy*

$$\frac{1}{\pi^2}\int_j^k \lambda = \frac{\kappa}{2}(a - w). \tag{12}$$

| $(j,k) \sim (j',k')$ | $d$ | $r$ | $D$ | $o$ | $a$ | $w$ | $\kappa$ | $\mathcal{A}/\pi^2$ |
|---|---|---|---|---|---|---|---|---|
| $(1,6) \sim (15,20)$ | 42 | pt | $(8,4)$ | $1 \to 6$ | $80/21$ | 2 | $1/2$ | $-19/42$ |
| $(1,7) \sim (14,20)$ | 42 | D | $(3,1)$ | $7 \to 1$ | $-8/7$ | $-2$ | 1 | $3/7$ |
| $(1,10) \sim (11,20)$ | 84 | E | $(6,3)$ | $1 \to 10$ | $124/7$ | 14 | $1/2$ | $-13/14$ |
| $(1,12) \sim (9,20)$ | 126 | A | $(5,2)$ | $1 \to 12$ | $288/7$ | 38 | $1/2$ | $-11/14$ |
| $(1,16) \sim (5,20)$ | 168 | G | $(3,1)$ | $1 \to 16$ | 36 | 34 | $1/2$ | $-1/2$ |
| $(2,7) \sim (14,19)$ | 42 | pt | $(8,4)$ | $7 \to 2$ | $24/7$ | 2 | $1/2$ | $5/14$ |
| $(2,10) \sim (11,19)$ | 84 | pt | $(4,2)$ | $2 \to 10$ | $96/7$ | 12 | 1 | $-6/7$ |
| $(2,12) \sim (9,19)$ | 126 | pt | $(5,3)$ | $2 \to 12$ | $192/7$ | 26 | 1 | $-5/7$ |
| $(2,17) \sim (4,19)$ | 168 | pt | $(3,1)$ | $17 \to 2$ | $244/7$ | 34 | $1/2$ | $3/14$ |
| $(3,4) \sim (17,18)$ | 0 | pt | $(10,5)$ | $4 \to 3$ | $-116/21$ | $-6$ | $1/2$ | $5/42$ |
| $(3,9) \sim (12,18)$ | 42 | pt | $(3,1)$ | $3 \to 9$ | $-16/21$ | $-2$ | 1 | $-13/21$ |
| $(4,5) \sim (16,17)$ | 0 | A | $(10,5)$ | $4 \to 5$ | $-36/7$ | $-6$ | $1/2$ | $-3/14$ |
| $(4,9) \sim (12,17)$ | 42 | C | $(8,4)$ | $4 \to 9$ | 8 | 6 | $1/2$ | $-1/2$ |
| $(4,11) \sim (10,17)$ | 84 | B | $(7,3)$ | $4 \to 11$ | $284/7$ | 38 | $1/2$ | $-9/14$ |
| $(4,13) \sim (8,17)$ | 126 | pt | $(6,2)$ | $4 \to 13$ | $1496/21$ | 70 | $1/2$ | $-13/42$ |
| $(5,11) \sim (10,16)$ | 84 | D | $(3,2)$ | $5 \to 11$ | $188/7$ | 26 | 1 | $-3/7$ |
| $(5,14) \sim (7,16)$ | 126 | E | $(6,2)$ | $14 \to 5$ | $488/7$ | 70 | $1/2$ | $-1/14$ |
| $(5,15) \sim (6,16)$ | 126 | pt | $(5,3)$ | $15 \to 5$ | $544/21$ | 26 | 1 | $-1/21$ |
| $(6,7) \sim (14,15)$ | 0 | pt | $(10,5)$ | $7 \to 6$ | $-212/21$ | $-10$ | $1/2$ | $-1/42$ |
| $(6,10) \sim (11,15)$ | 42 | pt | $(2,1)$ | $6 \to 10$ | $188/21$ | 8 | 1 | $-10/21$ |
| $(7,8) \sim (13,14)$ | 0 | pt | $(10,5)$ | $7 \to 8$ | $-28/3$ | $-10$ | $1/2$ | $-1/6$ |
| $(7,10) \sim (11,14)$ | 42 | F | $(9,4)$ | $7 \to 10$ | 32 | 30 | $1/2$ | $-1/2$ |
| $(7,12) \sim (9,14)$ | 84 | B | $(8,3)$ | $7 \to 12$ | $500/7$ | 70 | $1/2$ | $-5/14$ |
| $(8,10) \sim (11,13)$ | 42 | pt | $(1,1)$ | $8 \to 10$ | $68/3$ | 22 | 1 | $-1/3$ |
| $(8,12) \sim (9,13)$ | 84 | pt | $(2,2)$ | $8 \to 12$ | $932/21$ | 44 | 1 | $-4/21.$ |

*Thus* 42 *roots have negative limiting action and* 8 *have positive limiting action. Every limiting action is nonzero, and the minimum absolute value is $\pi^2/42$.*

*Proof.* For seams $j < k$, canonicalized output points agree precisely when their $n$-intervals overlap after a shift $d \in 42\mathbb{Z}$. Intersecting the twenty intervals in Lemma 4.1 gives 58 nonempty triples $(j,k,d)$. Twenty-eight contain one diagonal point strictly inside a vertical interval. The other thirty are circle-circle overlaps. Substitution of the certified normal roots from Lemma 4.2 leaves one simple root in 22 and none in the eight listed pairs. The strict angle boxes lie inside their overlap intervals.

At a diagonal-circle root, the limiting directions in scaled coordinates are $(1, 10/21)$ and $(0, \Delta n_j)$. At a circle-circle root in rows $i, i'$, the determinant has leading sign

$$\operatorname{sgn}(\Delta n_j \Delta n_k) \operatorname{sgn}\bigl(G_i'(\theta) - G_{i'}'(\theta)\bigr).$$

The rational derivative certificates therefore fix the degree-one ordering and prove transversality. Proposition 3.3 excludes additional off-seam roots, while Proposition 3.2 gives an embedded local resolution at each fold.

For an overlap $(j, k, d)$ choose a common height $h$ and put

$$p = (j, h/21), \qquad q = (k, (h+d)/21)$$

in $(\gamma/\pi, \theta/\pi)$ coordinates. Follow the singular path from $p$ to $q$. If $k - j$ is even, its endpoints differ by an ordinary torus deck transformation and $\kappa = 1$. If $k - j$ is odd, concatenate the path with its endpoint translate and divide the resulting integral by two; then $\kappa = 1/2$.

Concatenate this ordinary loop with its pillowcase-involuted image and close the remaining ends by mutually cancelling auxiliary arcs. Stokes' theorem and the residue $-\pi^2$ at each lattice point give

$$\frac{1}{\pi^2} \int_p^q \lambda = \frac{\kappa}{2} \left( \text{Area} - \sum_{z \in \mathbb{Z}^2} \text{wind}_z \right),$$

which is (12). Shoelace evaluation on the vertices of Lemma 4.1 gives the printed $a$ and $w$. The common height can move inside its overlap interval without crossing a lattice point, so the integral is constant and may be evaluated at a rational midpoint even when the actual normal root is algebraic.

For example, $(1, 7)$ has $(a, w, \kappa) = (-8/7, -2, 1)$, so the $7 \to 1$ action is $3\pi^2/7$. The pair $(3, 4)$ has $(a, w, \kappa) = (-116/21, -6, 1/2)$, so the $4 \to 3$ action is $5\pi^2/42$. The half-period map sends

$$(j, k, d) \longmapsto (21 - k, 21 - j, d)$$

and preserves the normalized action, ordering, and short deck class. Each row therefore has multiplicity two. Reading the last column proves the sign census and the minimum gap. □

**Corollary 4.4** (Small-perturbation persistence)**.** *There is $t_0 > 0$ such that, for every $0 < t < t_0$, the curve $L_t$ has exactly the 50 roots in Proposition 4.3, and all have the signs shown there.*

*Proof.* Transversality continues all 50 roots uniquely. Compact separation of the graph sheets and the fold neighborhoods excludes new roots. Proposition 2.3 gives primitives that vary continuously with $t$, and the canonical paths remain in the compact corner-free arc $I_0$. Thus every action converges to its table value. Since the finite minimum absolute limiting value is $\pi^2/42$, every sign is constant after decreasing $t_0$. □

*Proof of Theorem 1.1.* Exactness is Proposition 2.2, and the relative primitive is fixed by Proposition 2.3. Proposition 4.3 proves the complete root and limiting-action census. Corollary 4.4 proves persistence for small positive $t$. □

## 5. Energy interpretation and scope

The sign table has an elementary consequence for any exact holomorphic configuration that may be defined using $L_t$. The statement does not assume that a compactified figure-eight operation exists.

**Lemma 5.1** (Root-action identity)**.** *Let a finite tree of exact pseudoholomorphic or Morse components have no external inputs and one external root $x$ on $L_t$. Suppose its seam and boundary lifts are fixed and every component has the usual Stokes energy identity. Then*

$$E_{\mathrm{tot}} = \mathcal{A}_t(x).$$

*Consequently, a nonconstant such configuration can be rooted only at a positive-action root.*

*Proof.* Apply Stokes' theorem to each component. At every internal flag the same primitive difference appears with opposite signs on the two adjacent components. The action drops along Morse edges cancel in the same way. With no external inputs, only the external output remains. Compatible almost-complex and Morse data make each energy summand nonnegative, and a nonconstant component makes the total positive. □

**Corollary 5.2.** *For $0 < t < t_0$, every nonconstant exact zero-input configuration satisfying the hypotheses of Lemma 5.1 is rooted at one of the eight positive roots in Theorem 1.1.*

**Remark 5.3** (No counting conclusion)**.** Corollary 5.2 is an action obstruction, not an existence or parity theorem. It does not assert that any of the eight positive roots supports a holomorphic configuration, that only one vertex occurs, or that a sum of outputs defines a bounding cochain. Those conclusions would require compactification, transversality, gluing, local mod-two counts, and continuation data that are not constructed here.

**Remark 5.4** (Exact certificate)**.** The command

```
python3 pillowcase/q7_exact_actions.py
```

reconstructs the path word, verifies the rational root enclosures and derivative signs, enumerates all 58 overlaps, and prints the 25-row area-residue table using exact rational arithmetic. The standard-library program is an independent reproducibility certificate for the finite calculations, not a numerical premise of the proof.

Theorem 1.1 therefore supplies a complete finite action filtration for this composition. It leaves the holomorphic and categorical questions where they belong: outside the claimed result.

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